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One World, One Cut: A Real-Wave Interpretation of Quantum Measurement with a Located Boundary

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Abstract. We state an interpretation of quantum measurement and price it. Its commitments are four: the wave is real; there is one world; the Heisenberg cut is a physical crossover with a computable location, a computable width, and a physical address, the absorption vertex of a detector site; and outcome selection is a stochastic event at that vertex with the hazard of the golden rule, put in as a postulate. Exact calculations in the smallest models containing the relevant physics establish that the cut's location sits where the off-shell return rate balances the coupling's population-transfer rate, that its width is a layer of share K/ω equivalent to a crossover of relative width order one in the return-rate variable, that irreversibility requires a record channel dense on the capture timescale, and that neither the carrier frequency nor a self-sustaining amplifier sharpens the crossover. All of this is compatible with decoherence theory. What is distinctively the interpretation's own is stated as such: that recoverability ends in principle rather than for all practical purposes, that the crossing is coupling-set rather than mass-set or temperature-set, the one-world commitment with the nonlocality it costs, and a specification of what any mechanism for the selection event must supply. No new constant is introduced and no prediction differs from quantum mechanics.

1. Scope

Three processes, one dispute. Measurement talk merges three physically distinct processes. (1) *Entangling interaction*: system and apparatus couple; on the joint Hilbert space this is unitary evolution, linear and reversible, and nothing distinguishes a measurement from any other interaction — von Neumann's point, and the origin of the chain. (2) *Decoherence*: the apparatus is open to an environment; the reduced state obeys an irreversible master equation that is still linear in the density matrix, so a superposition becomes a mixture of its branches, weighted as before, never one of them. Irreversibility is purchased; selection is not. (3) *The event*: one outcome, in one world. Interpretations disagree about the third. This paper affirms that it occurs, places it at the absorption vertex of a detector site, and postulates it.

What kind of paper this is. An interpretation. It modifies no equation, introduces no constant, and makes no prediction that differs from quantum mechanics as used. What it offers is an account of what the equations describe and of where, physically, the description stops being reversible, together with an honest

statement of what that account costs. Its claims about the cut (§4) were tested in exact models before being stated; its claim about selection (§5) is a postulate, written in the form those tests said any mechanism would have to meet.

2. Relation to existing frameworks

Decoherence. The cut is the decoherence programme’s home ground, and everything in §4 is compatible with it. Zeh, Zurek, and Joos and Zeh established that the placement of the split is dynamical: the environment monitors the apparatus in a pointer basis it selects, and interference between pointer states becomes unobservable at a rate the coupling sets [Zeh 1970; Zurek 1981, 1982; Joos & Zeh 1985]. Tegmark made the consequence explicit as apparent collapse by scattering [Tegmark 1993]; Brune and colleagues measured a meter’s progressive decoherence as its states were made more distinguishable, a crossover of the kind §4 describes [Brune et al. 1996]; quantum Darwinism made the environment’s redundant records the criterion of objectivity [Ollivier et al. 2004; Zurek 2009]. The programme’s own limit is stated in Schlosshauer’s review: decoherence explains the appearance of a cut and not the occurrence of one outcome [Schlosshauer 2004]. Schlosshauer and Camilleri, asking whether decoherence challenges Bohr’s doctrine that a classical side must exist, conclude that it justifies where the cut may be placed without making the cut a physical discontinuity [Schlosshauer & Camilleri 2008; Camilleri & Schlosshauer 2015] — the “for all practical purposes” Bell would not accept [Bell 1990]. This paper accepts every einselection result, adds the one thing the programme concedes is missing, and concedes in turn that the addition is a postulate. Where it differs from decoherence-only is in two places: §4.5, where it argues that recoverability ends in principle and not FAPP, and §5, where it affirms one outcome.

Everett. Denies the event; every branch is real. Buys locality at the price of many worlds, with the Born weights supplied by decision-theoretic arguments whose adequacy is contested both ways [Everett 1957]. This paper makes the opposite trade (§6).

Spontaneous collapse. GRW, CSL, and the Diósi–Penrose models postulate the event as new stochastic dynamics with new constants [Ghirardi et al. 1986; Bassi et al. 2013; Diósi 1989; Penrose 1996]. Their natural parameter space has been largely removed by X-ray emission bounds and underground tests [Donadi et al. 2021], and they place the classical boundary on mass or size. This paper places it on coupling (§4.2), introduces no constant, and §7 states the dial that discriminates the two placements.

Bohm and Nelson. The family this paper belongs to is the single-world, ψ -ontic, nonlocal one, and Bohm and Nelson are its neighbours [Bohm 1952; Nelson 1966]. The distinction is where the probability measure sits: over hidden configurations of the propagating system in Bohm’s case; in a hazard at a detector site with a physical address, a tested location, and a computable width in this one — a measure about detectors, which laboratories can interrogate. Bohm’s foliation is empirically inaccessible by construction; this paper’s (§6) is carried openly and carries tests.

The transactional interpretation. The nearest neighbour. Cramer’s interpretation, built on Wheeler–Feynman absorber electrodynamics, describes a quantum event as a transaction between an emitter’s retarded offer wave and an absorber’s advanced confirmation wave, and reads the Born weight as their product $\psi\psi^*$ [Wheeler & Feynman 1945; Cramer 1986]. Cramer noted that the Dirac and Klein–Gordon equations supply the advanced solutions and treated the absorber functionally, without microscopic description. Kastner’s relativistic version adopts Davies’ direct-action electrodynamics with fermionic currents as sources and sets spin aside [Davies 1971, 1972; Kastner 2012]; Kastner and Cramer define an absorber as an atom or molecule and derive the Born rule for radiative processes as the offer–confirmation product [Kastner & Cramer 2018]; the absorber’s definition has been criticized as underspecified [Boisvert & Marchildon 2013]. Shared with this paper: actualization at the absorber, one world, nonlocality, and a square that comes from the structure of a rule rather than from a probability postulate. Different: this paper has no advanced waves; it locates the cut, gives it a width, and has tested the location; it places commitment at the vertex with a dense-record requirement; and it does not present the square as derived — $\psi\psi^*$ is a rule about products as the golden-rule hazard is a rule about rates, and in neither is the square explained.

Curie–Weiss models of registration. Allahverdyan, Balian, and Nieuwenhuizen solve the dynamics of the record’s amplification and assume selection at the crucial step [Allahverdyan et al. 2013]. This paper

agrees that registration is statistical mechanics and places the event one stage earlier (§4.3).

Copenhagen and QBism. Deny that the wave is a physical thing that evolves; the cut is a convention or a boundary of an agent’s knowledge. This paper takes the wave to be real and the cut to have an address, and pays for the realism in §3.6 and §6.

3. Ontology

3.1 Real waves. The wave is a physical field: the Dirac field for matter, the Maxwell field for light. “Particle” names a persisting excitation, and persistence is a property of bound states — the displaced pole of a dressed excitation — needing no mechanism beyond the equations. The virtual/real distinction is propagating versus evanescent: real wavenumber, phase advancing, energy carried away, versus imaginary wavenumber, exponentially damped, clinging to its source.

3.2 Superposition as coexistence. A superposition is both components present with a definite relative phase — coexistence, never alternation. When the components differ in energy their beat at the Bohr frequency is the laboratory signature that both are there: quantum beats, Ramsey fringes, and Schrödinger’s *zitterbewegung* for a Dirac wavepacket containing both energy signs [Schrödinger 1930]. On the Bloch sphere the free dynamics precesses the phase at fixed latitude; the outcome populations are constants of the unitary motion and alternation would destroy the Born weights. The latitude carries the weight; the move from latitude to pole is the event of §5.

3.3 The register. One two-sector structure recurs: the electron’s chiral pair, the photon’s helicity pair, the interferometer’s which-arm pair — an $SU(2)$ register storing its observable as a relative amplitude and phase, intrinsic in the first two cases and geometric in the third, where a beamsplitter sculpts the mode function into two branches. Path superposition is not a puzzle about a delocalized photon; it is the delocalization, shaped [Grangier et al. 1986]. Polarization is the helicity register: the Poincaré sphere is its Bloch sphere. The division the register enforces: delocalization is the wave’s birthright; discreteness is the detector’s budget — one quantum, one closure, however thinly the amplitude is spread.

3.4 The photon. Vacuum Maxwell in Riemann–Silberstein form splits into two decoupled helicity sectors [Bialynicki-Birula 1996]: the register with the off-diagonal coupling set to zero, no mass, no beat, no internal clock. Matter can lend the photon a coupling; everything clock-like about light is borrowed. A massive excitation, by contrast, carries a real phase at the Compton frequency, demonstrated interferometrically [Lan et al. 2013]; it is a phase, representation-dependent [Foldy & Wouthuysen 1950], and not a coupling.

3.5 Two classical limits. Matter becomes classical by being recorded, one absorption at a time [Tonomura et al. 1989]; Pauli exclusion bars a fermion field from macroscopic occupation of one mode, so there is no classical electron wave. Light becomes classical by occupancy, the coherent state [Glauber 1963], loud rather than averaged; thermal light is classical only as noise [Hanbury Brown & Twiss 1956]. Spin-statistics distributes the two limits. A single photon, too free to be recorded in flight and too small to be loud, is excluded from both, and at interaction is annihilated, converting itself into a record in matter.

3.6 One world. Exactly one outcome occurs. Bell’s theorem presents the bill at once: a single world with definite outcomes that reproduces the quantum correlations is nonlocal. The trade is made in §6.

4. The cut, with an address

Each claim is cited to the exact calculation that bears on it (`heisenberg_cut_recoverability/` in the repository), run with predictions fixed before the results were opened.

4.1 The criterion. The cut is where *in-principle recoverability ends*: the first point in the chain at which no operation, however idealized, can restore the pre-event configuration. Operationally, recoverability is the partial Loschmidt echo — the site’s own Hamiltonian reversed for the capture time, the environment untouched. It is not the reversal an experimenter can apply to the site–field coupling alone: even with no environment, flipping the coupling returns a detuned capture only on resonance, because the detuning keeps running forward; the echo returns it exactly at every detuning. Experiments that locate the cut by a reversal protocol must separate the two.

4.2 Location and width. A site holding energy $e < E_0$ is off-shell with deficit ΔE , and the deficit is a return rate, $\kappa_{\text{ret}} = \Delta E/\hbar$, at which the virtual excitation returns to the field. The cut sits where the return rate balances the coupling’s population-transfer rate, $\kappa_{\text{ret}}/K = 1$. The layer within which the site’s restoring rate is below the coupling has share width $w = K/\omega$; in the return-rate variable the same layer is a crossover of relative width order one. Under the passive-absorber proxy $K \sim \Gamma$ the width is tabulable: about 10^{-8} for an alkali line, 10^{-18} for an optical clock transition, 10^{-7} – 10^{-6} for a transmon, 10^{-3} – 10^{-2} for a room-temperature silicon photodiode [Steck 2025; Ludlow et al. 2015; Kjaergaard et al. 2020; Sabbah & Riffe 2002]. Superconducting circuits are macroscopic by particle count yet behave as clean two-level systems because their w is atomic-grade: mass and particle number are not the variable; κ_{ret}/K is.

exact-model result	value	consequence
rate-based crossover midpoint, linear absorber with a 256-mode record channel	$\kappa_{\text{ret}}/K = 1.3\text{--}2.0$ across two decades of Γt	the location claim holds, with K the population-transfer rate; a fixed-observation-time reading drifts outward, so the location is a statement about rates
relative width of the crossover in κ_{ret}/K	25–75 % points at 1.2–3	the share layer K/ω , in the other variable
carrier frequency, rotating-wave approximation removed	centre shifts by $-(1.1\text{--}1.5)K^2/\omega$; half-width unchanged to 2 % for $\omega/K \geq 8$	ω enters as the Bloch–Siegert shift and in the share conversion, nowhere else
record channel of 1, 4, 16, 64, 256 modes	coherent exchange; no leak; recurrence; convergence only at 256	a record is a cut only where it is a continuum on the capture timescale
self-sustaining amplifier swept through its gain	stored energy at the field-set value through both Hopf points (slope 0.004), rising only past the running onset (slope 1.1)	the amplifier kinks the recoverability crossover; it does not sharpen it

4.3 Three stages, one crossing.

stage	dynamics	recoverability
capture	unitary, distributed across candidate sites in proportion to local intensity	recoverable — demonstrated by spin echoes, photon echoes, atomic-frequency-comb memories, and the catch-and-reverse of a monitored quantum jump [Hahn 1950; Kurnit et al. 1964; de Riedmatten et al. 2008; Afzelius et al. 2009; Mineev et al. 2019]
commitment	the absorption vertex and thermalization at one site: in a semiconductor the interband vertex, 10 fs–1 ps	in-principle recoverability ends here
registration	reservoir-powered amplification	irrecoverable and macroscopic; photon-agnostic; no which-site weight

Commitment is the vertex, not the record. Everything downstream is a deterministic cascade that copies what the vertex decided and contributes no statistics; the commonplace that collapse happens when the record is amplified misassigns the cut by one stage. Cooling widens the reversibility window without moving

the crossing. Two physical requirements: the record channel must be dense on the capture timescale (one record degree of freedom gives coherent exchange, not irreversibility); and the amplifier’s engaged regime — a free phase exists but is entrained — leaves recoverability where the field set it, the energy bookkeeping changing hands from field to site only at the onset of free running, continuously.

4.4 The detector taxonomy. The photon-counting detector is the projective limit: one quantum, one vertex, one click, the only class for which per-event Born claims are properly made. NMR/MRI is the ensemble limit, with no single-quantum threshold. The cloud chamber is the repeated limit, a sequence of capture–commit–register triplets [Mott 1929]. The photographic emulsion is the integrating limit [Gurney & Mott 1938], and its reciprocity failure — a latent speck decaying if not reinforced — is everyday evidence that commitment and registration are physically distinct. Every click deposits the full quantum at one site; distributed capture is reversible energy, never banked; losing sites retain no residue.

4.5 What recoverability is, and the one claim that is this paper’s own. A reversal requires the stored phase history to point back along the path. Whether it survives is set by the reference structure of the medium: references *common* across sites (a hologram, a homodyne local oscillator) give an invertible record; references *different but deterministically known* (an inhomogeneous ensemble) give a record invertible by rephasing, which is why echoes work; references *random and unrecorded* (a thermal detector bulk) give a record that cannot be inverted. The third case is made by heat: thermal motion delivers phase kicks at random, unrecorded times, the record of the incident wave’s history spreads over e^{S/k_B} configurations of the medium, and Landauer’s bound prices the reversal at $T\Delta S$, which for a thermalized photon is the photon’s own energy [Landauer 1961]. Polarization echoes exhibit the strong form, in which past a point the decay is set by the medium’s own many-body dynamics and a better reversal does not help [Levstein et al. 1998; Pastawski et al. 2000].

Here is the claim decoherence theory does not make, stated as an argument. Decoherence says the interference terms survive in the global state and are inaccessible for all practical purposes. This paper says that when the references are random and unrecorded, no agent — not merely no practical agent — holds what a reversal requires, and recoverability ends *in principle* at the vertex. The argument’s weakness is plain: e^{S/k_B} is astronomical, not infinite, and a Poincaré recurrence is not forbidden. We hold the claim, flag it, and note that everything else in this section stands without it.

5. Selection at absorption

5.1 The postulate. A quantum arrives at a detector as a wave spread over many sites. Three things then happen, and the paper postulates all three.

1. *Every site draws, in proportion to the square.* A detector site at the single-photon level is a linear absorber: it has no phase of its own, the wave creates its oscillation, and the phase difference between the two is fixed by the site’s detuning and return rate. The energy the site draws from the field is proportional to the square of the local amplitude, weighted by a Lorentzian in its detuning of width set by the coupling. Eligibility is graded, not sharp; no site is excluded.
2. *A sink opens at one site, by chance.* At each site a sink can open at any moment, with a probability per unit time proportional to the energy that site has drawn so far, the same rule at every site; the first sink to open anywhere is the one that opens. *By chance*: not when the site’s phase reaches alignment — that rule makes the winner the fastest of N nearly deterministic approaches, whose statistics are wrong. *Drawn so far*: the rate is linear in that energy; a square-root or quadratic rate gives the wrong statistics, and the rate’s timing does not matter.
3. *The whole quantum flows through it.* Once a sink is open, the quantum’s entire energy passes through that one site, and the amplitude at every other site vanishes — including sites in the other arm of an interferometer, metres away — within a thousandth of the time the sink took to open, faster than light crosses the detector. One sink per quantum.

Formally: commitment at the vertex is a stochastic event whose hazard at a site is linear in the site’s absorbed energy, hence proportional to the squared amplitude, with a site-independent time profile; exactly one site completes; and the constraint enforcing “exactly one” acts across the detector faster than light. Clause 1 is the linear response, the golden rule’s matrix element squared; clause 2 is Fermi’s golden rule read as an

event rather than a rate [Dirac 1927; Fermi 1950]; clause 3 is the nonlocality that one world costs. The paper derives none of the three.

5.2 What any mechanism must supply. The form of the postulate is not arbitrary. An attempt to derive it from a synchronization dynamics in a real-wave substrate — a race among detector-site oscillators locked to the field — was tested exhaustively before this paper was written, and it is recorded in the repository as a negative result (`NEGATIVE_RESULT.md`; `adler_two_channel_exploratory/RESULTS.md`). Its residue is a specification. A first-to-align rule gives an outcome exponent of 1.5 in the coupling, with the site count entering only as its logarithm. A memoryless hazard linear in absorbed energy gives 2.1–2.2, the Born value plus a tenth of a power that belongs to the sharp eligibility threshold of a self-sustaining oscillator and does not go away with grid, window, pulse, or noise; the graded weight of clause 1 has no threshold and no residual. The hazard’s form is load-bearing (a square-root hazard gives 1.7, a quadratic one 3.2); its timing is not. And exclusivity cannot be local: the one-quantum stop must act within about 10^{-3} of the commitment time to reproduce single-photon coincidence data, and an ensemble reading of independent site rates is excluded by two orders of magnitude. Any successor mechanism must meet each clause.

5.3 Relation to the transactional picture, exactly. Both put actualization at the absorber and get the square from the structure of a rule. The transactional interpretation’s rule is a product of forward and backward waves; this paper’s is a hazard linear in absorbed energy. Neither derives the square, and neither predicts anything quantum mechanics does not. What this paper adds is the address: a location for the cut that has been tested, a width, a dense-record requirement, and a timescale audit placing commitment at the vertex.

6. Entanglement and the price of one world

(Conditional on the real-wave commitment extended to the non-separable two-particle configuration; the configuration-space objection — that the many-particle wave lives on $\mathbb{R}^{3N}$ — lives here and is held open.)

Two objects. The local thermal reference at each detector — the random, unrecorded references that make commitment possible — is established in the common past of both wings and is independent of the analyzer settings: it is a Bell common cause, and by Bell’s theorem it cannot source the correlations [Bell 1964]. The correlation comes from the second object: the non-separable field configuration carrying the entangled two-particle mode across both wings — one configuration of the real field spanning A and B. Locating the nonlocality there adds no postulate; it restates the real-wave commitment. Settings are free; no-signalling is not assumed but follows from the hazard’s linearity in the squared amplitude, which is Gisin’s argument run in reverse as a consistency condition — a nonlinear hazard would signal [Gisin 1990].

The trade. Everett buys locality with many worlds; this paper buys one world with nonlocality in Bell’s sense, never signalling. No interpretation pays neither price.

The foliation, priced. Nonlocal commitments must be ordered, and the ordering is a preferred slicing that enters through the measurement sector only: capture is exactly Lorentz covariant, and the frame appears where the event does. As written this is explicit low-energy Lorentz violation confined to measurement. Ordinary matter couples to it, beyond the electromagnetic commitment itself, only gravitationally, which is why clock-comparison and Michelson–Morley bounds have not registered it; the quantitative demonstration that the residual anisotropy clears every established bound is owed. Whether the frame is local or cosmic is open and testable (§7).

7. Consequences and tests

Every test here is either a consistency test that quantum mechanics also passes or a test of one of this paper’s own premises.

The coupling dial. The cut is placed on κ_{ret}/K ; the collapse programs place theirs on mass. An engineered system moved along the coupling dial at fixed mass discriminates the two: a transmon coupled to a lossy readout mode, where the two rates are independently engineered, swept through the coupling ratio, recording

commitment latencies and the echo-recoverable window against a circuit-QED null model. Being decoherence-compatible, the framework curve coincides with the null; the dial discriminates this interpretation from mass-based collapse, not from decoherence. Macromolecule interferometry reads accordingly: interference is lost where the coupling budget says, and the fullerene thermal-emission result already reads that way [Hackermüller et al. 2004].

Temperature as a dial on the record channel. Temperature is a proxy for the rate $\Gamma(T)$ of unrecorded phase kicks. The exact model gives recoverability $R = \exp[-2\Gamma t f(\Delta)]$ with f the site’s occupation, independent of the channel’s bandwidth at fixed Γ ; hence retrieval efficiency versus storage time at several temperatures collapses onto one curve in $\Gamma(T)t$ if the channel is memoryless, and the retrieval crossover in input detuning sits at $|\Delta| \approx 2K$ and does not move with temperature while $\Gamma(T) \ll K$. A rare-earth photon-echo memory at variable input detuning and three or four temperatures tests both; the rare-earth coherence laws are tabulated [Equall et al. 1995; Könz et al. 2003]. Both predictions coincide with Markovian decoherence theory; the interpretation owes them and cannot be discriminated by them.

The frame’s tests. Because the foliation enters through measurement only, its tests are boundary tests in the dissipation channel, which existing free-precession bounds do not probe: sidereal modulation of multipartite correlations and of interferometric quantum noise, and surface relaxation in magnetic resonance. Sidereal nulls retire a cosmic frame.

What falsifies what. A mass-located boundary on the coupling dial falsifies this paper; a coupling-located one falsifies mass-based collapse. A linear absorber with a dense record channel whose recoverability crossover sat far from $\kappa_{\text{ret}}/K = 1$ would have falsified the location in the simplest model; it sat at 1.3–2.0. Curvature in the conditional statistics of a photon split between two ports of mismatched structure — which any local POVM makes affine in the splitting ratio — would overturn the linear structure of quantum mechanics itself; its null closes a channel and leaves the core intact.

8. Open problems

1. **A quantum self-sustaining absorber with a record channel.** The exact tests used a linear absorber and, for the amplifier, a classical oscillator with gain by hand. Showing the slaved, engaged, and running regimes with κ_{ret} a coherent return rather than a damping is an open-system calculation not yet done.
2. **In principle versus for all practical purposes.** §4.5’s claim is an argument; making it a result requires showing that a random-unrecorded reference structure forbids reversal by any agent, or conceding the point.
3. **The covariant formulation** of the selection event on the foliation, with its no-signalling proof independent of the statistics it reproduces, and the clearance of the Lorentz bounds.
4. **The configuration-space objection**, held open at the non-separable configuration: a premise priced, not a problem solved.

9. Conclusion

Schrödinger built an equation that describes the evolution of the wave and meant it as physics; the description was extended to relativity and to fields and then stopped, one postulate short. This paper does not remove the postulate. It gives it an address, a width, a time, and a form: the wave is real and evolves; where it is absorbed, one site commits, in one world, at a place the physics of the detector fixes and a time the vertex fixes; the cut is there, with a width the coupling sets; and what the postulate does not explain is stated as a postulate, priced, and handed forward with the specification any successor must meet.

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